Ravi Prakash Srivastava

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Research Interests

Large Language Models (LLMs), AI Agents, Trustworthy AI, NLP, Adversarial Machine Learning, Computer Vision.

Education

Indian Institute of Information Technology Ranchi (IIIT Ranchi), India *August 2024 – May 2026*
M.Tech (Data Science & AI) *CGPA: 8.79/10*
Institute of National Importance, Govt.of India

North Eastern Hill University (NEHU), Shillong, India *August 2018 – June 2022*
B.Tech (Information Technology) *CGPA: 7.85/10*
Central University, Govt.of India

Experience

Software Engineer *SS Software Solutions LLC,Hyderabad,India* *March 2026 – Present*

- Defined and enforced ML pipeline standards encompassing model versioning (MLflow/DVC), structured logging (JSON schema-validated with trace IDs for distributed tracing), and observability instrumentation (metrics, alerts, drift detection), reducing incident diagnosis time by 40% and improving pipeline reliability across 3+ production models.
- Standardized ML pipeline governance across versioning, logging, and monitoring layers, cutting untracked model deployments to zero and reducing mean-time-to-debug (MTTD) by ∼35%; applied STRIDE-based Threat Modeling to identify 8+ design-level risks including data poisoning vectors and model inversion attack surfaces.
- Automated security testing by integrating SAST tools (Bandit, Semgrep) and dependency vulnerability scanning into CI/CD pipelines (GitHub Actions), reducing manual security review effort by ∼60% and accelerating release cycles.

Machine Learning Intern *C-DAC (Govt. of India),Hyderabad,India* *August 2022 – February 2023*

- Performed large-scale data preprocessing and feature engineering on student activity logs (attendance, assignments, quizzes) to support predictive modeling.
- Implemented and benchmarked multiple machine learning models, including Decision Trees, Random Forests, and Support Vector Machines, for academic performance prediction.
- Evaluated model performance using cross-validation and metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC, identifying the best-performing model for early prediction of at-risk students.

Research Projects

- **Adversarial Prompt Injection Attacks on LLM: Cryptographic Key Leakage and Defense Strategies**
  Under Publication at International Conference on Advanced Computing and Applications (ICACA-2026).
- **IoT-HITS: IoT-based Human Intrusion Detection System for Border Regions using Deep Learning**
  [IETE Journal of Research] DOI: `https://doi.org/10.1080/03772063.2025.2521688`
- **Deepfake Video Detection Using Face-Centric Processing and Frame Sampling**
  Under Publication International Conference on Advanced Computing and Applications (ICACA-2026).
- **Balanced Few-Shot Episodic Learning for Accurate Retinal Disease Diagnosis**
  (arXiv: `https://arxiv.org/abs/2512.04967v1`)
- **Smart City Vehicle Accident Monitoring and Detection System Using MEMS, GSM, GPS Raspberry Pi 4**
  [IETE Journal of Research] DOI: `https://doi.org/10.1080/03772063.2022.2043787`

Projects

Cross-Domain Text Summarization Using Transformer Models *Master's Thesis*

- Investigated domain shift and catastrophic forgetting by adapting **BART-large-CNN** from news summarization to legal documents using the **BillSum** dataset.
- Implemented parameter-efficient fine-tuning (**LoRA**) with **Elastic Weight Consolidation (EWC)** to preserve pre-trained knowledge during domain adaptation.
- Achieved improved cross-domain generalization while fine-tuning only **1–2%** (~2M of 406M) of model parameters, significantly reducing GPU memory usage and training cost.

AI-Powered Vulnerability Management Platform (SAST + DAST) `Live Demo`

- Developed an AI-powered vulnerability management platform integrating Static Application Security Testing (SAST) and Dynamic Application Security Testing (DAST).
- Automated static code analysis using **Semgrep** for early detection of security vulnerabilities and code quality issues.
- Built an interactive **Streamlit** dashboard for real-time vulnerability visualization, severity classification, and remediation recommendations.

LLM Semantic Cache & Analytics Dashboard `Live Demo`

- Designed and implemented an embedding-based semantic caching layer for Large Language Models to retrieve responses using semantic similarity instead of exact keyword matching.
- Reduced LLM API latency and inference costs through intelligent cache reuse while maintaining response quality.
- Developed a real-time analytics dashboard to monitor cache hit/miss rates, similarity thresholds, query latency, and overall system performance.

Technical Skills

- **Technologies:** Python, JavaScript.
- **Databases:** SQL, MariaDB, MySQL, relational schema design.
- **Machine Learning & AI:** Deep Learning, Machine Learning, Computer Vision, Image Processing, Deepfake Detection, Large Language Models (LLMs), Hugging Face Transformers, PyTorch, Scikit-learn, OpenCV.
- **MLOps & DevOps:** Git, Docker, GitHub Actions (CI/CD), MLflow, DVC.

Achievements

- Received the **Institution of Electronics and Telecommunication Engineers (IETE) Scholarship** for academic excellence during M.Tech. studies in Computer Science and Engineering.
- Received the **Best Paper Award** at **ICACA 2026** for the paper, *Adversarial Prompt Injection Attacks on Large Language Models: Cryptographic Key Leakage and Defense Strategies.*

References

- **Dr. Ajay Kumar**
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- **Prof. Md. Iftekhar Hussain**
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